



Loan Compliance Platform

Agentic AI system for automated loan compliance & review

TEAM RED PANDAS

Ranajay Sit · Raunak Gupta · Deepayan Choudhury · Toju George · Tapas Kumar



The Problem

Banks must validate thousands of loans before reporting them — yet today's process is slow, manual, and opaque.

Current Process

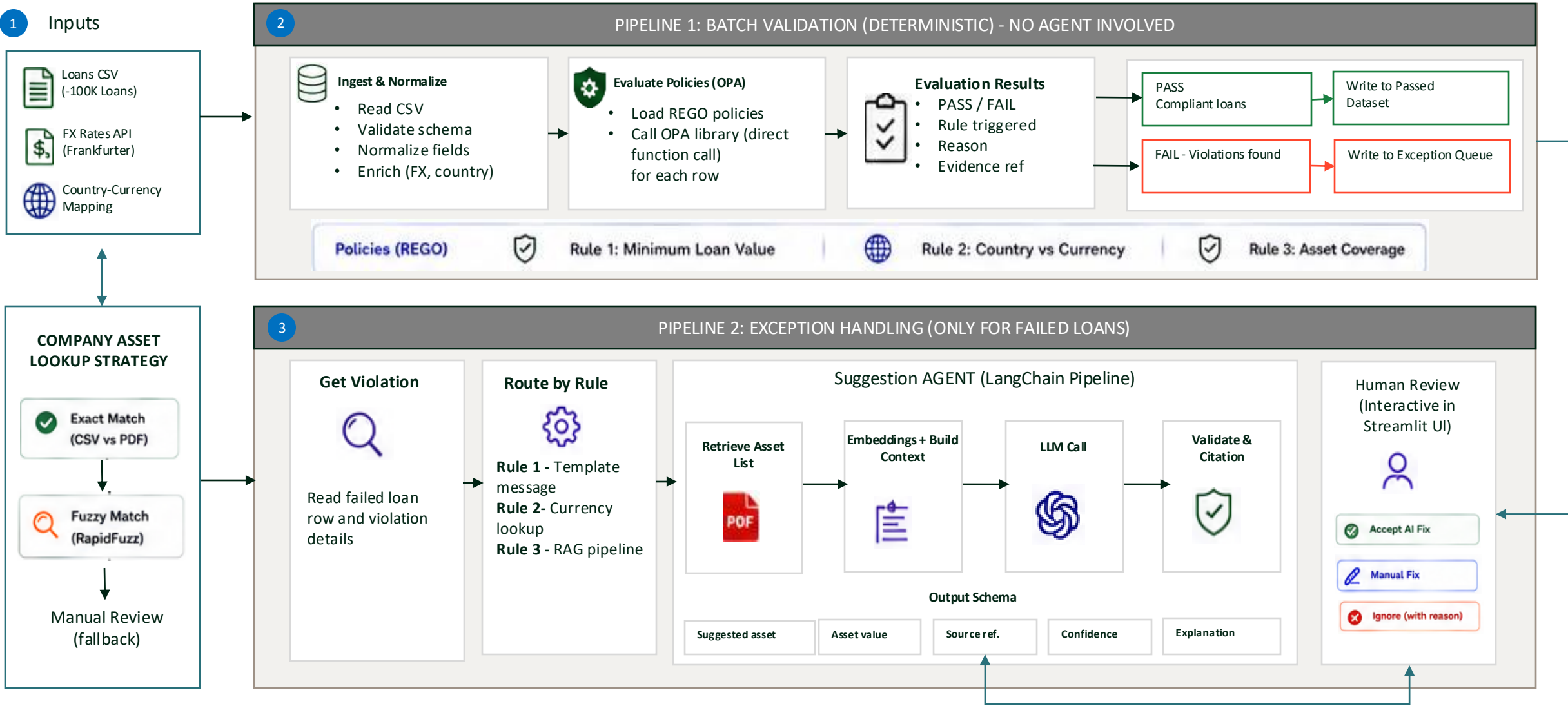
- Manual compliance checking is **slow and repetitive**
- Reviewers need **explanations**, not just pass/fail results
- Some failures require **searching company documentation** for possible fixes

Our Goal

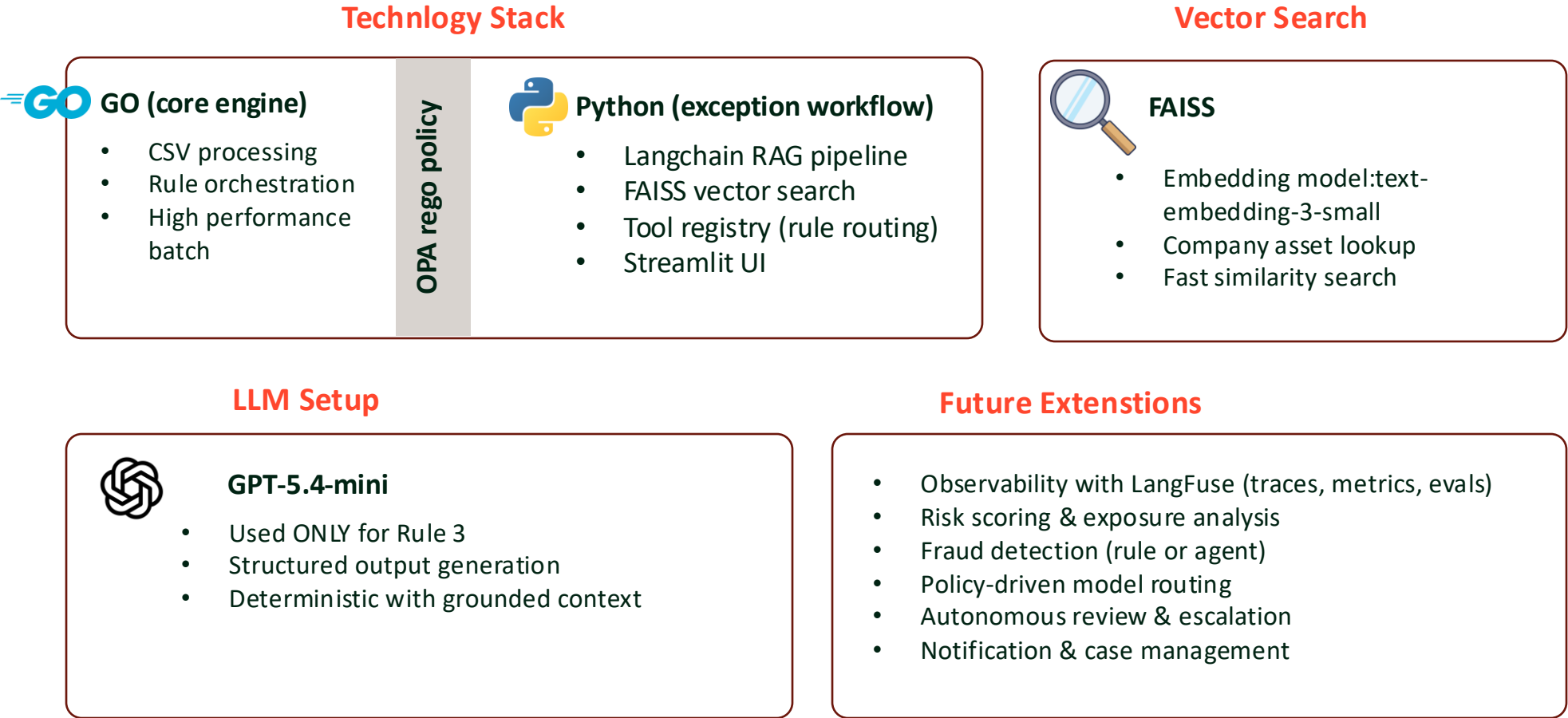
- Automate compliance checks
- Suggest meaningful fixes using AI
- Keep a human reviewer in the loop
- Generate an overall compliance report

**Rule Engine decides → AI explains & suggests
→ Human approves**

High Level Architecture



High Level Architecture



Engineering Decisions & Why They Matter

Design Decision	Why We Built It	Criteria Addressed
OPA + Rego for rules	Deterministic, auditable compliance	Rule engine, explainability
Cached FX rates	One API call per run, consistent results. The FX rate on the date when the pipeline was run is considered for the scenario.	Low latency, better accuracy
LLM + Embedding Model	Generates a structured response, keeps it deterministic	Good contextual understanding, allowing it to interpret retrieved company information and propose valid asset substitutions.
FAISS	Integrates well with langchain and eliminates need for DB creation	efficient local vector search for ~100 company documents.
RAG only for Rule 3	Retrieval needed only for asset replacement	Appropriate AI usage
Paginated review UI	Handles 100K+ loans efficiently	Scalability, Simple to use

kyndryl.

LIVE DEMO

kyndryl.

Thank You